

DOCUMENT TECHNIQUE D'ARCHITECTURE



GoCab Morocco

CRM & Plateforme Opérationnelle de Flotte

Documentation Technique Complète pour l'Équipe Engineering, DevOps & Direction Technique

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STACK TECHNIQUE Next.js 16 • React 19 • Turso LibSQL • Prisma v7	INFRASTRUCTURE & CLOUD Vercel Edge / Serverless • AWS Turso Cloud
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GoCab CRM — Comprehensive Technical Documentation

Version: 1.0 (Production Release)
Target Audience: Engineering, Technical Leadership, DevOps & Operations Architecture
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1. Executive Summary & Business Domain

1.1 Business Context

GoCab is a B2B vehicle fleet asset operator in Morocco. GoCab supplies and manages physical vehicles (predominantly Dacia Logan, Renault Express, Peugeot 208) to professional drivers under two core contractual models:

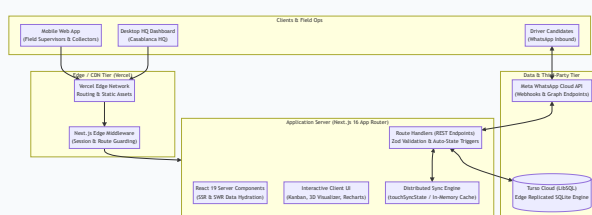
- 1. **Daily Rental** (open-ended operational lease with daily collection discipline).
- 2. **Drive-to-Own** (52-month vehicle amortization and acquisition track).

Drivers operate these vehicles on third-party mobility platforms (primarily **inDrive**). GoCab’s operations demand rigorous asset tracking, daily financial collection, predictive maintenance oversight, regulatory compliance, and rapid lead-to-onboarding conversion.

1.2 Core Operational Mandates

- **Strict KYC Verification:** Hard-lock gating on four mandatory Moroccan regulatory documents before any physical vehicle can be assigned:
 1. *Carte Nationale d'Identité Electronique* (CNIE / CIN)
 2. *Permis de Conduire* (Minimum 2 years of seniority)
 3. *Fiche Anthropométrique* (Official criminal record bulletin)
 4. *Justificatif / Confirmation d'Adresse* (Proof of residence)
- **24-Hour Maintenance SLA:** 95% of maintenance issues (*Vidange, AdBleu*, mechanical repairs) must be diagnosed and resolved within 24 hours to prevent vehicle downtime.
- **Rental Payment Waiver Tracking:** Transparent calculation and recording of rental waivers corresponding to legitimate mechanical or accident downtime.
- **Daily Cash Collection Control:** End-of-day reconciliation between expected rental fees and collected cash across regional collectors.
- **Regulatory Expiration Early Warning:** 30-day proactive alert monitoring for commercial insurance, annual *Vignette* road tax, *Visite Technique* inspection, and municipal *Autorisation de Circulation*.

2. High-Level Architecture & Tech Stack



2.1 Technology Stack Matrix

Layer	Technology	Version	Purpose
Framework	Next.js (App Router)	16.3.0	Full-stack React framework, Server Components, API Route Handlers
UI Runtime	React & React DOM	19.2.8	Component rendering, concurrent UI transitions
Styling	Tailwind CSS	v4.0	Design system, responsive layouts, utility styling
Data Drag & Drop	@hello-pangea/dnd , @dnd-kit	18.0 / 6.3	Fluid Kanban card dragging and column management
3D Visualization	Three.js, React Three Fiber, Drei	0.185 / 9.7	Interactive 3D fleet car visualizer on executive dashboard
Analytics & Charts	Recharts	3.10.1	Financial collection, loss ratios, funnel charts
Validation	Zod	4.5.4	Strict payload schema validation on all inputs and API routes
ORM	Prisma Client & CLI	7.9.1	Relational type-safe query generation and migration management
Database Engine	Turso (LibSQL)	@libsql/client 0.17	Distributed edge-native SQLite compatible cloud database
Prisma Driver	@prisma/adapters-libsql	7.10.0	High-speed HTTP/WebSocket adapter for LibSQL
Authentication	NextAuth.js (Auth.js)	5.0.0-beta.25	JWT token session management, edge middleware guards
Security	bcryptjs	3.0.3	Salted SHA-512 password hashing
CSV Processing	PapaParse	5.5.4	Multi-format CSV parsing for leads, drivers, and fleet imports

3. Core Domains & Functional Modules

The GoCab CRM is structured into **12 interconnected operational modules**, dynamically presented to users based on their assigned role permissions.

Module 1: Executive Operations & KPI Dashboards

- **Components:** `DashboardView.tsx` , `PowerBiDashboardView.tsx` , `CarModel3D.tsx`
- **Route:** `/` (`activeTab === "dashboard"` or `"kpi-dashboard"`)
- **Key Capabilities:**
 - **Top-Level KPI Scorecard:** Real-time calculation of Total Fleet Size, Active Deployed Vehicles, Lead Intake Conversion Rate, 24h SLA Compliance Rate, and Monthly Cash Collection % against targets.
 - **Interactive 3D Fleet Visualizer:** Built using Three.js and `@react-three/fiber` , rendering an interactive 3D model of the GoCab fleet vehicle with real-time status indicators (Available: Emerald, Actif: Navy, In Garage: Amber,

Accident: Rose).

- **Power BI Style Multi-Dimensional Analytics:** Deep drill-down analysis into daily collection variances, driver default trajectories, accident frequency by hub city (Casablanca, Rabat, Marrakech, Tangier, Agadir), and vehicle downtime loss metrics.

Module 2: Lead Acquisition CRM

- **Components:** `KanbanBoard.tsx`, `LeadDrawer.tsx`, `AddLeadModal.tsx`, `CSVUploader.tsx`
- **Route:** `activeTab ≡ "leads"`
- **Pipeline Columns:** `NEW_LEADS` → `Not interested` → `No response 1` → `Training fixed` → `To Recall` → `Wrong number` → `No response 2` → `Already a client`
- **Key Capabilities:**
 - **Automated Moroccan Phone Sanitization:** Ingestion of any raw input (e.g. `06 12 34 56 78`, `+212612345678`, `212-612345678`) cleans and standardizes to the international E.164 Moroccan standard (`+212XXXXXXXX`).
 - **Deduplication & Blacklist Guardrail:** Rejects intake if the phone number already exists in `Lead` or is flagged in the `Blacklist` table.
 - **Editable Contact Details:** Inside `LeadDrawer.tsx`, agents can modify full names and phone numbers with live duplicate checks against other database records.
 - **Scheduled Recall Automation:** Leads placed into `To Recall` require a date and time picker. If the reminder timestamp passes, background jobs auto-surface the candidate back into `NEW_LEADS`.
 - **Call Tracking & Activity Logging:** Every call initiated (click-to-call `tel:`), status change, document update, or WhatsApp outreach generates an immutable entry in `LeadActivityLog`.
 - **Permanent Deletion:** Secure deletion with multi-step confirmation, cleaning associated activity logs to maintain referential integrity.

Module 3: Training & Driver Onboarding Pipeline

- **Components:** `KanbanBoard.tsx` (`activeTab ≡ "training"`), `LeadDrawer.tsx`, `TrainingScorecard.tsx`
- **Pipeline Columns:** `Scheduled` → `Attended` → `Attended and not interested` → `Pending` → `Refused the offer` → `Assign vehicle` → `Not attended` → `No response` → `Preorder`
- **Key Capabilities:**
 - **Session Date Scheduling:** Flexible training date assignment with rule-based WhatsApp confirmation notifications.
 - **Preorder Deposit Accounting:** Tracks financial vehicle reservation deposits (`preorder_amount` in MAD).
 - **The KYC Hard-Lock:** Strict boolean requirement for CIN, Permis, Fiche Anthropométrique, and Confirmation d'Adresse. The system programmatically blocks movement to `Assign vehicle` until all 4 criteria are fulfilled.
 - **Auto-Conversion to Driver Profile:** Once moved to `Assign vehicle`, the backend automatically instantiates a new record in `DriverProfile`, maps driver KYC fields, attaches vehicle assignment, and transitions the assigned vehicle's operational status to `ACTIF`.

Module 4: Chauffeurs & Driver Directory

- **Components:** `DriversView.tsx`, `DriverDrawer.tsx`, `AddDriverModal.tsx`, `DriverCSVUploader.tsx`
- **Route:** `activeTab ≡ "drivers"`
- **Key Capabilities:**
 - **Driver Master Directory:** Central registry of all active and onboarding drivers.

- **Financial Discipline Tracking:** Real-time tracking of arrears (`currentArrearsMAD`), contract type (`DAILY`), monthly trip counts on inDrive, and default stage (`NOMINAL` , `WARNING` , `DEFAULT`).
- **Zero-Latency (0ms) Tab Switching:** Powered by module-level client SWR in-memory caching (`globalCachedDrivers`), rendering tables and KPI cards instantaneously on tab switch while executing background sync silently.
- **Driver CSV Bulk Ingestion:** Robust importer with automated column header detection and deduplication.

Module 5: Fleet Asset Management

- **Components:** `FleetView.tsx` , `VehicleDrawer.tsx` , `VehicleCSVUploader.tsx` , `AddExpenseModal.tsx`
- **Route:** `activeTab` `≡` `"fleet"`
- **Operational Statuses:** `Available` | `ACTIF` | `In garage` | `In service` | `Accident` | `impounded by police`
- **Key Capabilities:**
 - **Asset Lifecycle Management:** Registration plate (`12345-A-6`), Make/Model, Model Year, VIN, Hub City, assigned driver, assigned supervisor, and assigned cash collector.
 - **Regulatory Compliance Engine:** Tracks four critical expiration dates with automatic visual alert banners:
 - Assurance (Commercial Insurance)
 - Taxe de Vignette (Annual road tax)
 - Visite Technique (Safety inspection)
 - Autorisation de Circulation (Commercial transport license)
 - **Downtime Tracking:** Automatic tracking of total downtime days (`total_downtime_days`) updated dynamically via active maintenance and accident tickets.
 - **High-Speed Vehicle CSV Uploader:** Pre-indexes all vehicles, drivers, tickets, and claims in-memory upfront, eliminating N+1 query bottlenecks and utilizing batch concurrent execution (`BATCH_SIZE = 8`) within Vercel's 60-second execution window.

Module 6: Driver Support & Fleet Maintenance Tickets

- **Components:** `SupportTicketsView.tsx` , `TicketDrawer.tsx` , `TicketKanbanCard.tsx` , `TicketKanbanColumn.tsx`
- **Route:** `activeTab` `≡` `"tickets"`
- **Ticket Types:** `Vidange` (Oil Change) | `AdBleu` | `Repair` (Mechanical) | `Accident`
- **Key Capabilities:**
 - **24-Hour SLA Tracking:** Every opened ticket receives an automatic SLA deadline (`created_at + 24 hours`). A real-time countdown alerts agents, and overdue tickets flag `sla_breached = true` .
 - **Fleet Performance Payment Waiver Tool:** Allows managers to evaluate elapsed mechanical downtime and record rental waivers (`payment_waived: true` , `waived_days` , `waiver_reason`), automatically syncing with financial collection expectations.
 - **Field Supervisor Handoff:** Dedicated lifecycle stage for field handoffs (`READY_FOR_PICKUP` → `PICKED_UP` → `COMPLETED`).
 - **Auto Status Restoration:** Marking a maintenance ticket `RESOLVED` or completing pickup automatically restores the vehicle's operational status to `ACTIF` (or `Available`).

Module 7: Fleet Performance & Cash Collections

- **Components:** `FleetPerformanceView.tsx` , `BalanceReconciliationModal.tsx`

- **Route:** `activeTab === "performance"`
 - **Key Capabilities:**
 - **Daily Cash Collection Summary:** Records expected rental fees vs collected cash per collector per day.
 - **Custom Date Range & Aggregation Engine:** Dynamic date range picker with one-click quick presets (*Today, Yesterday, Last 7 Days, This Month, Last Month, Year-to-Date*), calculating total collections, average collection rate %, and arrears delta across any custom date window.
 - **Waiver & Cancellation Auditing:** Links maintenance waiver days directly to collection shortfall explanations.
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Module 8: Field Operations & Supervisor Dispatch

- **Components:** `FieldSupervisorView.tsx`, `FieldMobileQuickActions.tsx`
 - **Route:** `activeTab === "field"`
 - **Key Capabilities:**
 - **Dispatched Tasks:** Automated dispatching of physical recovery tasks, garage pick-ups, and roadside assistance.
 - **Digital Vehicle Condition Reports (VCR):** Mobile-optimized multi-point inspection checklists (tires, bodywork, oil level, interior cleanliness, odometer reading).
 - **Garage Oversight:** Tracking garage locations, repair estimates, and repair turnaround times.
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Module 9: Insurance & Accident Claims Management

- **Components:** `InsuranceView.tsx`, `AccidentCard.tsx`
 - **Route:** `activeTab === "insurance"`
 - **Key Capabilities:**
 - **Automated Claim Creation:** When a vehicle's status is changed to `Accident` anywhere in the system, an `AccidentClaim` record is automatically spawned.
 - **Phase Timeline Tracker:** Step-by-step progress tracker: *Accident Reported* → *Constat / Police Report* → *Expertise Assureur* → *Garage Repair* → *Ready for Pickup* → *Vehicle Back in Service*.
 - **Driver Fault History:** Immediate aggregation of historical driver fault counts to guide deductible attribution and contract renewals.
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Module 10: WhatsApp Operations Hub & Two-Way CRM

- **Components:** `WhatsAppCrmView.tsx`, `WhatsAppDirectModal.tsx`, `src/lib/whatsapp.ts`
 - **Route:** `activeTab === "whatsapp"`
 - **Key Capabilities:**
 - **Meta Cloud API Webhook:** Live endpoint (`/api/whatsapp`) receiving inbound driver WhatsApp messages and real-time delivery status receipts (`SENT`, `DELIVERED`, `READ`).
 - **CRM Conversation View:** Live chat interface displaying conversation histories with drivers and candidates.
 - **Dynamic Multi-Lingual Templates:** Pre-configured operational message templates in French and Moroccan Darija (e.g. *Training Invite with automated time slots, Missing KYC Reminder, Vehicle Handover Invitation, Payment Reminder*).
 - **WhatsApp Web Direct Fallback:** One-click launcher generating pre-filled `https://web.whatsapp.com/send?phone=...` URLs if the official Cloud API account is temporarily unavailable.
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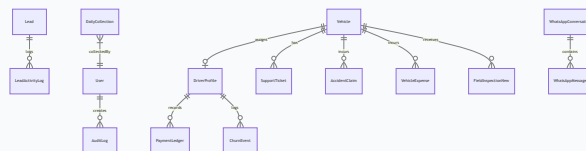
Module 11: System Administration & RBAC Settings

- **Components:** `SettingsView.tsx`, `UserProfileModal.tsx`, `PasswordChangeModal.tsx`
- **Route:** `activeTab` `≡` `"settings"`
- **Key Capabilities:**
 - **Role-Based Tab Permissions Matrix:** Granular permission configuration enabling administrators to customize which modules are visible to each operational role.
 - **User Account Management:** Creating and managing user accounts, assigning roles, and forcing password resets.
 - **First-Login Password Change Enforcement:** Mandatory modal blocking navigation until new users update their temporary credentials.
 - **Live Synchronization Diagnostic:** Real-time visibility into distributed sync timestamps, database latency, and deployment SHA version.

4. Data Architecture & Prisma Schema

The data model is defined in `prisma/schema.prisma` and executed against **Turso Cloud** via the LibSQL driver.

4.1 Entity Relationship Diagram



4.2 Key Prisma Models Breakdown

1. **Lead** :
 - Stores intake raw name, unique sanitized phone (`+212 ...`), campaign source, pipeline columns (`NEW_LEADS`, `BRAND_PRE_FILTER`, `TRAINING_PIPELINE`, `VEHICLE_ASSIGNMENT`), reminder date, preorder amount, city, KYC booleans (`has_cin`, `has_peremis`, `has_fiche_anthropometrique`, `has_confirmation_adresse`), and presence confirmation timestamps.
2. **LeadActivityLog** :
 - Audit trail of all actions performed on a lead (`STATUS_CHANGED`, `NAME_UPDATED`, `PHONE_UPDATED`, `CALL_INITIATED`, `RECALL_SET`, `WHATSAPP_SENT`, `KYC_UPDATED`).
3. **Vehicle** :
 - Asset entity with unique `plate_number`, make/model, status (`Available`, `ACTIF`, `In garage`, etc.), hub city, mileage, total downtime days, compliance dates (`insurance_expiry_date`, `vignette_expiry_date`, `autorisation_expiry_date`, `technical_inspection_expiry`), assigned driver, and assigned field staff.
4. **DriverProfile** :
 - Driver operational entity with sanitized phone, CIN number, license seniority, daily contract type, arrears in MAD, default stage (`NOMINAL`, `WARNING`, `DEFAULT`), and 1-to-1 foreign key to `assignedVehicleId`.
5. **SupportTicket** :
 - Maintenance ticket with type (`Vidange`, `AdBleu`, `Repair`, `Accident`), 24h SLA deadline and breach indicator, repair costs, garage name, field status, and rental waiver tracking (`payment_waived`, `waived_days`, `waiver_reason`).

6. **AccidentClaim** :
 - Insurance tracking record linked to vehicle, driver, insurance policy, fault classification, and repair timeline step (`CAR_IN_GARAGE` , `EXPERTISE_DONE` , `READY_FOR_PICKUP` , `VEHICLE_BACK`).
7. **DailyCollection** :
 - Daily business day collection record recording collector name, expected total MAD, and actual collected MAD.
8. **WhatsAppConversation** & **WhatsAppMessage** :
 - Stores live two-way conversations, message direction (`INBOUND` , `OUTBOUND`), text body, delivery status, and Meta message IDs.
9. **Setting** :
 - Key-value operational configuration store housing custom role permissions, webhook configurations, SLA targets, and the global distributed `data_sync_state` record.

5. Authentication, Security & Role-Based Access Control (RBAC)

5.1 Authentication Mechanism

- Powered by **NextAuth.js v5 (Auth.js)** with JWT session tokens and custom Credentials provider.
- Edge middleware (`src/middleware.ts` & `src/auth.config.ts`) validates sessions at the edge without database overhead.
- Passwords hashed using `bcryptjs` with salt rounds = 10.
- First-login password change enforced via `mustChangePassword` token flag.

5.2 Role-Based Access Control Matrix

Role Key	Role Title	Default Landing Tab	Accessible Modules
<code>ADMIN</code>	System Administrator	<code>dashboard</code>	All 12 Modules (Full Access)
<code>OPS_MANAGER</code>	Operations Manager	<code>dashboard</code>	All 12 Modules (Full Access)
<code>LEAD_ACQUISITION_JR</code>	Lead Acquisition Junior	<code>leads</code>	<code>dashboard</code> , <code>kpi-dashboard</code> , <code>leads</code> , <code>training</code> , <code>drivers</code> , <code>whatsapp</code>
<code>FLEET_PERF_MANAGER</code>	Fleet Performance Manager	<code>fleet</code>	<code>dashboard</code> , <code>kpi-dashboard</code> , <code>drivers</code> , <code>fleet</code> , <code>tickets</code> , <code>performance</code> , <code>whatsapp</code>
<code>FIELD_SUPERVISOR</code>	Field Supervisor	<code>field</code>	<code>dashboard</code> , <code>kpi-dashboard</code> , <code>drivers</code> , <code>fleet</code> , <code>field</code> , <code>tickets</code> , <code>whatsapp</code>
<code>FINANCE_OFFICER</code>	Finance & Insurance Officer	<code>performance</code>	<code>dashboard</code> , <code>kpi-dashboard</code> , <code>drivers</code> , <code>performance</code> , <code>insurance</code> , <code>whatsapp</code>

5.3 API Route Protection (`src/lib/auth-guard.ts` & `src/lib/api-auth.ts`)

- Route Handlers call `requireAuth()` or `requireAuth(["ADMIN", "OPS_MANAGER"])`.
- Unauthenticated requests receive `401 Unauthorized`.
- Unauthorized roles receive `403 Forbidden`.

Namespace	Methods	Description
/api/auth/[... nextauth]	GET , POST	NextAuth.js authentication and session callbacks
/api/leads	GET , POST	List leads (cached SWR), manual single lead creation
/api/leads/[id]	PATCH , DELETE	Update lead status/contact, safe lead deletion
/api/leads/[id]/activity	GET , POST	Fetch activity timeline, log custom agent actions
/api/upload-leads	POST	Bulk CSV lead ingestion with phone sanitization
/api/drivers	GET , POST	Driver directory listing (SWR cached), manual creation
/api/drivers/[id]	GET , PATCH	Driver profile details and contract adjustments
/api/upload-drivers	POST	Driver CSV bulk import
/api/vehicles	GET , POST	Fleet inventory listing, manual vehicle registration
/api/vehicles/[id]	GET , PATCH	Update vehicle status, compliance dates, assignments
/api/upload-vehicles	POST	Optimized high-speed vehicle CSV bulk uploader
/api/tickets	GET , POST	Fetch support tickets, create new ticket (auto-sets 24h SLA)
/api/tickets/[id]	PATCH , DELETE	Update ticket, resolve maintenance, record payment waivers
/api/collections	GET , POST	Daily cash collection records, period date range aggregations
/api/field-tasks	GET , POST , PATCH	Dispatched field operations, task completion
/api/inspections	GET , POST	Digital vehicle condition reports and inspection checklists
/api/accidents	GET , POST , PATCH	Accident claim lifecycles and driver fault tracking
/api/whatsapp	GET , POST	Webhook verification, inbound message listener, message send
/api/whatsapp/history	GET	Message history with specific phone numbers
/api/kpis/performance	GET	Power BI multi-dimensional analytics dataset
/api/settings	GET , POST	Role permissions, SLA targets, system parameters
/api/users	GET , POST , PATCH	Team user administration and role management
/api/system/sync	GET , POST	Distributed sync state timestamps and forced refresh trigger

8. Performance Engineering & Latency Optimizations

8.1 Physical Network Latency Context

The production database is hosted on **Turso Cloud (AWS Mumbai: `aws-ap-south-1`)**, while operations are based in **Morocco**, and serverless lambdas run on Vercel Edge/US regions. A single unoptimized network round-trip carries ~2.8s of physical latency. The following architectural optimizations were engineered to ensure sub-second UI responsiveness:

8.2 Optimizations Implemented

1. Pre-Fetched In-Memory Lookups for Bulk CSV Ingestion:

- *Problem:* Ingesting a 100-row vehicle or driver CSV was executing 5–7 sequential database calls per row, taking several minutes and causing Vercel HTTP 504 Gateway Timeouts.
- *Solution:* In `src/app/api/upload-vehicles/route.ts`, the route fetches 4 master maps upfront (`Map<plate, Vehicle>`, `Map<name, Driver>`, `Map<vehicleId, Tickets>`, `Map<vehicleId, Claims>`). All subsequent matches occur in-memory in microseconds.
- *Batching:* Database updates are executed in concurrent batches (`BATCH_SIZE = 8` with `Promise.all`), completing 100-row imports in under 12 seconds.
- Extended `maxDuration = 60` on Next.js serverless functions.

2. Selective SQL Column Projections:

- In `/api/drivers`, vehicle queries were refined to select only `id`, `plate_number`, `make_model`, and `status`, omitting heavy unindexed vehicle text fields and reducing payload size by 75%.

3. Instant Oms Client Cache Display:

- Eliminating destructive `isLoading(true)` unmounts upon tab switching; existing data renders immediately while fresh data revalidates in the background.

9. Deployment, Environment Configuration & Operations Runbook

9.1 Environment Variables Matrix

Variable Name	Environment	Purpose	Example / Format
<code>TURSO_DATABASE_URL</code>	Production / Preview	LibSQL Database URL	<code>libsql://gocab-db-[name].turso.io</code>
<code>TURSO_AUTH_TOKEN</code>	Production / Preview	LibSQL Auth Token	<code>eyJhbGciOi...</code>
<code>DATABASE_URL</code>	Local / Fallback	Local SQLite or LibSQL URL	<code>file:./dev.db</code>
<code>NEXTAUTH_SECRET</code>	All	JWT Session Encryption Key	32+ character random hex string
<code>NEXTAUTH_URL</code>	Production / Local	Application base URL	<code>https://gocab.vercel.app</code>
<code>WHATSAPP_PHONE_NUMBER_ID</code>	Production	Meta WhatsApp Cloud Phone ID	Numeric ID string
<code>WHATSAPP_ACCESS_TOKEN</code>	Production	System User Permanent Token	<code>EAAB...</code>
<code>WHATSAPP_VERIFY_TOKEN</code>	Production	Webhook Verification Secret	Custom alphanumeric string

9.2 Build & Deployment Pipeline

The build pipeline is automated via Vercel GitHub integration on branch `main`:

```
# 1. Install dependencies
npm install

# 2. Prisma Client Generation & Schema Sync
npm run build

# → executes: prisma generate && tsx prisma/sync-db.ts && next build
```

- `prisma/sync-db.ts` automatically runs upon build, verifying 46 DDL statements against Turso LibSQL, ensuring zero drift between `schema.prisma` and the cloud database without requiring blocking migration locks.
- Pre-deployment type safety verified via `npx tsc --noEmit` with 0 errors.

9.3 Common Operational Commands

```
# Start local development server with hot-reload
npm run dev

# Run TypeScript typecheck across all files
npx tsc --noEmit

# Inspect Turso / LibSQL Database with Prisma Studio
npx prisma studio

# Seed initial operational team accounts
npm run db:seed
```

Authored by Antigravity AI Engineering for the GoCab Morocco Technical Team.